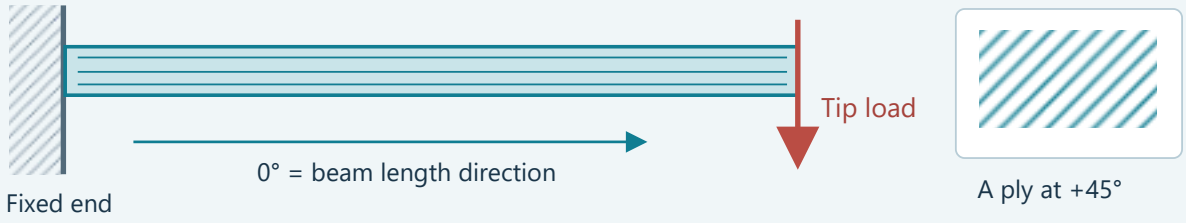
Composite optimization: how the two modes work

A plain-language guide to fully stressed and equal-stress orientation optimization, the implementation, and a worked example.

The shared goal: find a stiff laminate under the supports and loads defined in the imported PrePoMax model. At this stage, only the ply angles change.

Same beam, supports and load — change only the fiber angles



Fixed: ply count + thickness **Variable: angle of each ply**

Schematic top view of the planar beam; dimensions and deformation are not to scale.

Why rotating the fibers changes stiffness

A unidirectional ply is usually much stiffer along its fibers than across them. Rotating a ply changes how it carries the applied loading. The best arrangement depends on the entire model: its geometry, supports and load directions.

What can change?

The fiber angle of each independent ply. With symmetry enabled, mirrored plies change together.

What stays fixed?

Ply count, individual ply thickness, geometry, supports and loads. Thickness and ply-count optimization are later stages.

How the code measures stiffness

For a fixed applied force pattern, it minimizes **compliance**: how much the forces move the structure in their own directions.

$$C = F^T u = 2U$$

F = applied force vector; u = displacement vector; U = elastic strain energy.

For one force acting through displacement δ, **C = Fδ**. Under the same force, a smaller displacement means lower compliance and greater stiffness.

This explanation uses force control, as in the beam demo. For supported prescribed-displacement cases, the code instead maximizes 2U through the objective 1/(2U).

Reading order: page 2 — criteria; page 3 — simple example; page 4 — code; page 5 — your actual beam.

Same stiffness goal. An extra balance requirement.

Meaning of "fully stressed" in this project. Classical fully stressed design seeks efficient use of material, with components approaching their allowable stress under at least one load case. For laminates, this commonly involves thickness as well as orientation [1]. The current fixed-size option is an **orientation-only adaptation**; it does not enforce full utilization.

1. Fully stressed mode

Find the stiffest orientation arrangement that passes the strength constraint.

$$\text{Minimize } C$$
$$\text{with } \max(r_p) \leq r_{limit}$$

A stiff design can be accepted even if its plies are far below the stress limit.

2. Equal-stress mode

Find the stiffest arrangement that passes strength **and** balances ply utilization.

$$\text{Minimize } C$$
$$\text{with } \max(r_p) \leq r_{limit}$$
$$\text{and } CV(r) \leq \text{tolerance}$$

The default tolerance is 0.10, or 10% of the mean utilization.

What is ply stress utilization?

Utilization compares stress with the material's allowable strength. For example, **80 MPa / 100 MPa = 0.8 = 80%**. A ratio above 1 means that allowable is exceeded.

Local stress component	Ratio checked by the code
Along fibers, σ11	|σ11| / Xt in tension; |σ11| / Xc in compression
Across fibers, σ22	|σ22| / Yt in tension; |σ22| / Yc in compression
In-plane shear, τ12	|τ12| / S

Xt, Xc, Yt, Yc and S are positive material strengths. For each ply p, r_p is the largest of these ratios over all its integration points. This is an in-plane maximum-stress check.

What does "equal" mean here?

It means similar **peak normalized utilization between plies**. Longitudinal and transverse raw stresses need not match, because their allowable strengths differ.

$$CV = \text{population standard deviation}(r_p) \;/\; \text{mean}(r_p)$$

CV = 0 means all ply utilizations match. CV ≤ 0.10 means their standard deviation is at most 10% of their mean; it does not impose a 10% maximum pairwise difference.

Equal-stress mode keeps minimizing compliance once the balance requirement is satisfied. It may need to accept a softer layup to meet that requirement.

02 / AN EASY WORKED EXAMPLE

Why the modes can choose different layups

Imagine a two-ply beam with a fixed thickness and a 10 N tip force. The FEM solver returns the following results for three trial angle arrangements.

Illustrative numbers: this teaching example is invented to show the selection rules. The actual eight-ply beam results appear on page 5.

Layup A

Stiff, uneven utilization

Ply 1	90%
Ply 2	30%

Strength passes; balance fails

Layup B

Slightly softer, balanced

Ply 1	62%
Ply 2	60%

Strength and balance pass

Layup C

Stiffest, but overstressed

Ply 1	110%
Ply 2	80%

Strength fails

Trial	Deflection δ	C = 10 N × δ	Ply utilizations	CV
A	1.0 mm	10 N·mm	90%, 30%	50%
B	1.2 mm	12 N·mm	62%, 60%	1.64%
C	0.8 mm	8 N·mm	110%, 80%	15.8%

Use a strength-utilization limit of **1.0 (100%)** and an equal-stress tolerance of **0.10 (10%)**.

Fully stressed → A

A and B both pass strength. A has the lower compliance: 10 versus 12 N·mm. It is the stiffer feasible choice.

Equal stress → B

A fails the CV limit. B passes strength and balance, so it is selected despite its slightly greater deflection.

C fails both modes. Although it has the lowest compliance, its first ply is at 110% utilization and exceeds the strength limit.

Check the CV calculation by hand

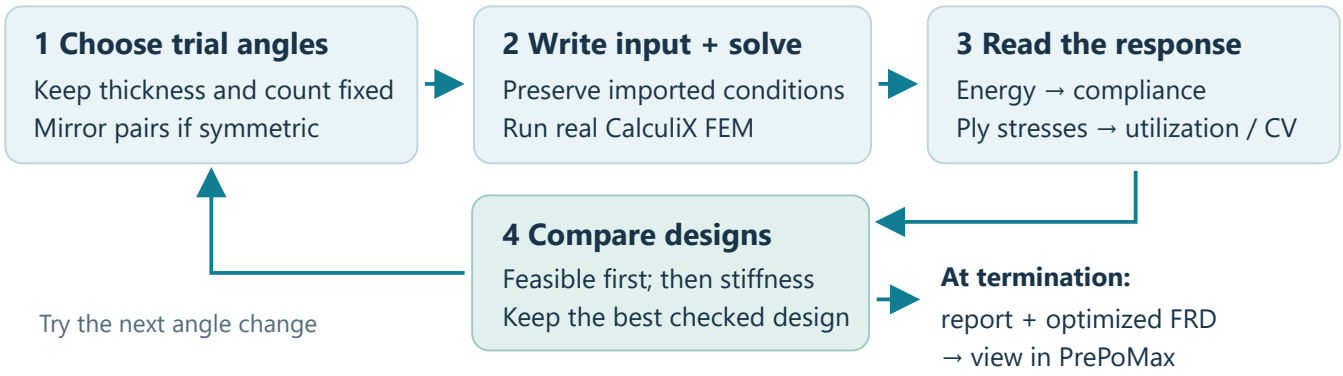
For A, mean = (0.90 + 0.30)/2 = **0.60**. The population standard deviation is **0.30**, so CV = 0.30/0.60 = **0.50 = 50%**.

For B, mean = **0.61** and standard deviation = **0.01**. Therefore CV = 0.01/0.61 ≈ **0.0164 = 1.64%**, comfortably below 10%.

The strength rule controls the worst ply. The balance rule controls how unevenly the plies use their strength.

How the implementation works

Both modes use the same deterministic angle-search engine. The selected method changes the constraints and the order in which plies are considered.



Function and source location	Responsibility
`CalculixDesignEvaluator.__call__()` structural_response.py:173	Replace the laminate definition, write a candidate .inp, run CalculiX and read energy, motion and local ply stresses.
`stress_metrics()` structural_response.py:133	Calculate each ply's peak utilization, the overall maximum and the CV.
`constraint_violation()` orientation_optimization.py:49	Check strength; add the CV violation only for equal-stress mode.
`optimize_orientations()` orientation_optimization.py:55	Choose candidate angles, rank their FEM results and retain the best checked design.

How new angles are chosen

1. **Check starting designs:** the user's layup, then uniform 0°, 90°, +45° and −45° stacks.
2. **Change one independent angle at a time:** try positive and negative changes, initially 45°. Symmetric pairs move together.
3. **Use stress to choose the order:** fully stressed mode prioritizes high-utilization plies; equal-stress mode prioritizes utilization outliers.
4. **Refine the search:** reduce the angular move when a sweep finds no improvement, down to the selected resolution (default 1°).

Angles are snapped to the manufacturing grid. Equivalent 180° directions and repeated designs are cached, so the same angle arrangement is not solved twice.

> **How a winner is selected:** feasible designs rank first. Among feasible designs, lower compliance wins. While all candidates are infeasible, lower total constraint violation wins, followed by compliance.

The run stops at the resolution condition, evaluation budget, cancellation or a solver error. If no design meets the constraints, the result is marked infeasible. Resolution termination is a local search condition; global optimality is not guaranteed.

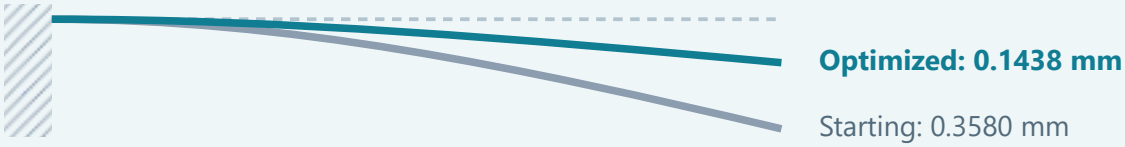
Compliance comes from integrated whole-model strain energy: CalculiX ELSE output with TOTALS=ONLY [2]. Per-ply constraints use local integration-point DAT stresses. The final FRD is the field result viewed in PrePoMax.

Source root: `D:\Git_Composite_Automation\fem-automation`.

What happened in the real run?

The 100 × 10 mm planar beam has a clamped end and a 9 N total transverse tip load **in its plane**. Both modes kept **8 plies × 0.25 mm = 2 mm** and found the same all-0° laminate.

Under the same 9 N total load, the beam bends less



Schematic deformation, visually magnified. Values are actual maximum reference-node displacements.

2.4901×
effective stiffness relative to start

+149.01%
effective stiffness gain

53
real FEM evaluations per mode

Quantity	Starting layup	Optimized layup
Angles	[45/−45/90/0]s	[0/0/0/0]s
Compliance	3.2131 N·mm	1.29035 N·mm
Maximum displacement	0.3580 mm	0.1438 mm
Maximum ply utilization	7.096%	1.778%
CV of ply utilizations	27.020%	0.0371%

"s" means mirror the listed half-stack. Stiffness ratio = C_{start}/C_{best}. The all-0° winner was checked as an early seed; 53 evaluations include the alternatives tested afterward.

Why the same answer? Fibers along the beam length gave the best stiffness found, and the resulting CV was already well below the 10% limit. The equal-stress requirement did not change the winning layup.

> **This beam is not classically fully stressed.** Its optimized maximum utilization is only about 1.78%. With sizing fixed, the run improves stiffness without removing material to raise utilization toward the limit.

Where to inspect the model and results

Native PrePoMax demo model
`D:\DataBin_Composite_Automation\cantilever_beam\cantilever_prepo.pmx`

Index of validated solver results
`D:\DataBin_Composite_Automation\CURRENT_RESULTS.json`

CURRENT_RESULTS.json lists the validated report.json, optimized.inp and prepomax_viewer\optimized.frd for both modes. The companion Desktop file **FEM_Automation_User_Instructions.pdf** explains how to open them.

Scope: planar shells and one linear-elastic static step. The in-plane stress criterion does not assess buckling, delamination or progressive damage. An all-0° optimum for this demonstration does not establish the best laminate for other loading conditions.

[1] Ghiasi et al., Optimum stacking sequence design of composite materials, Part II, 2010, §4.3.

[2] CalculiX 2.22 manual, *EL PRINT. Implementation notes: `docs\optimization_methods.md` in the project folder.